

Humayan Kabir



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Profile

Machine learning and data engineer with three years of experience in various production infrastructures. Most recently built Mercato, a multimodal product search engine with a multilingual architecture that currently supports both English and Japanese. Based in Dhaka, Bangladesh, open to relocating to Japan.

Skills and Tools Used

- **Search and Retrieval:**
 - BM25, FAISS, HNSW, bge-m3, CLIP.
 - Rank fusion, LambdaMART, LightGBM.
 - NDCG, Significance testing.
- **Machine Learning and AI:**
 - Python, PyTorch, Transformers.
 - GAN, CNNs, scikit-learn, Deep learning.
 - Computer Vision, Symbolic Regression.
 - Model Evaluation, Leakage Control.
- **Data and Backend:**
 - Apache Spark, Apache Airflow, HDFS.
 - PostgreSQL, MySQL, MongoDB.
 - FastAPI, Django REST.
- **Infrastructure:**
 - Google Cloud Run, Hugging Face Spaces.
 - Docker, Git, Unix, Shell Scripting.
- **Monitoring and Visualization:**
 - Zabbix, Grafana.
 - High-availability and parallel-processing architecture.

Blog

- Mercato: Multilingual Product Search
- The Round Wall: SHA-256 Cryptanalysis
- Kavel: Marketplace Listing Assistant
- humayankabir-re.github.io/blog

Language

- English – Fluent
- Bengali – Native

Professional Experience

Reliance Legal Services – AI Solutions Engineer (Contractual)

Dhaka, Bangladesh (Oct 2025 – Present)

- Designed and deployed an agentic RAG pipeline, enabling semantic search across historical briefs, contracts, and memos.
- Built an LLM-powered document generation system, reducing per-case formatting time from **30** minutes to under **1** minute.

Banglalink Digital Communication Ltd – Specialist OSS Developer

Dhaka, Bangladesh (Jan 2023 – Jun 2024) · via Searchlite Communications Ltd

- Led deployment and tuning of the high-availability monitoring platform. CPU usage down over **20%**, storage down about **50%**, alert-processing overhead down about **15%**.
- Onboarded 3 new engineers through hands-on project work.

Banglalink Digital Communication Ltd – Big Data Analyst

Dhaka, Bangladesh (Oct 2022 – Dec 2022) · via Searchlite Communications Ltd

- Built Spark and Airflow pipelines delivering stakeholder KPIs.
- Evaluated distributed ETL tooling across HDFS, Spark, Airflow, and tested knowledge with peers.

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Dhaka, Bangladesh (Jan 2022 – Sep 2022)

- Built and shipped containerised Django REST applications.
- Mentored 2 junior developers and business staff, aiding in effective in-house tool adoption.

Independent AI Projects (July 2024 – Present)

Mercato: Product Search Engine ([Google Cloud Run](#), [Hugging Face Spaces](#), [GitHub](#)).

- Three retrieval channels (BM25 keyword, bge-m3 dense, CLIP image), merged by rank fusion and reordered by a per-locale LambdaMART model.
- Evaluated on the full Amazon ESCI benchmark. 8,956 English and **10,407** Japanese queries.
- Image channel lifted NDCG@10 from 0.4545 to 0.4821, the reranker added 0.0319. HNSW cut median query time from 132 ms to **0.31** ms.

Neural Cryptanalysis of SHA-256

- Measured how far Neural Networks can predict SHA-256 output. Found it at the hardest stage of SHA-256 at the **4th** round out of 64 repeating rounds. Then it drops to random chance.
- Confirmed the method itself works by reproducing a published distinguisher on a weaker cipher (Speck32/64, **77%** accuracy at five rounds).

Education

North South University, Dhaka, Bangladesh (Jan 2022)

- Bachelor's in Computer Science and Engineering
- Majored in Machine Learning